

Multiplexing and Spreading

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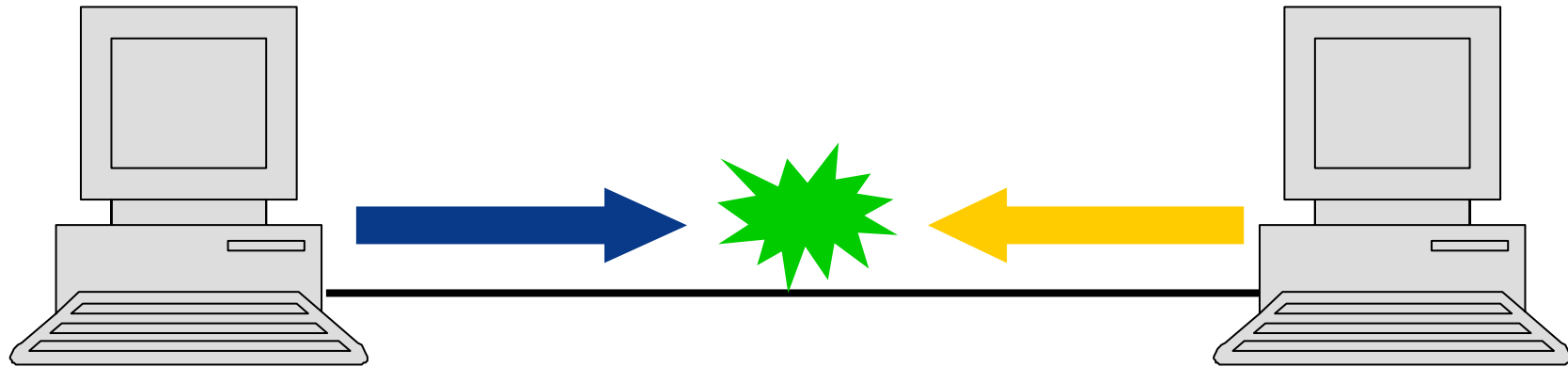
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Book chapters

Forouzan: 1.1, 6, 12.3

Kihl: 4.6, 3.3, 6.2 (Statistisk multiplexering)

Need for multiplexing



Two stations that transmit data on a physical link cannot do this simultaneously on the same frequencies with the same coding scheme.

Data flow

⌘ Simplex mode:

- ◆ The communication is unidirectional.

⌘ Half duplex mode:

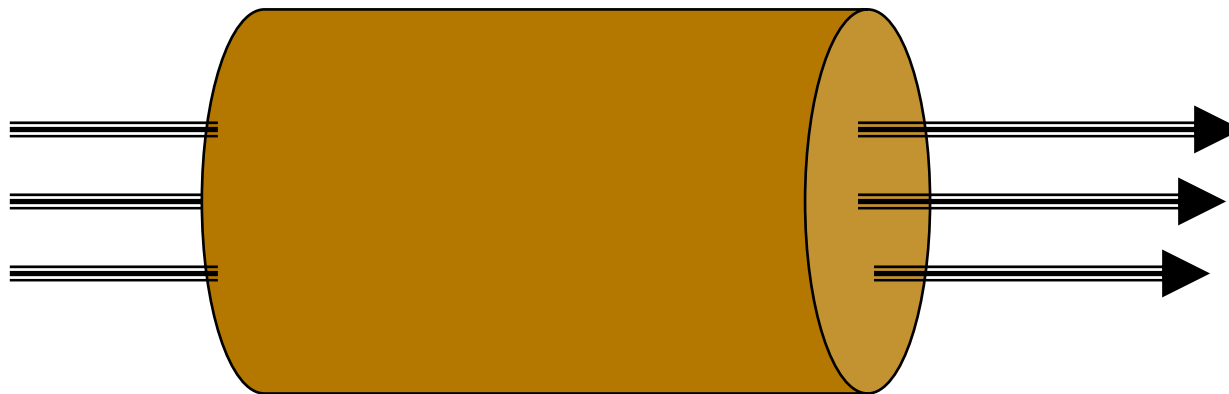
- ◆ Each station can both transmit and receive, but not at the same time.

⌘ Full duplex mode:

- ◆ Both stations can transmit and receive simultaneously.
- ◆ Requires that the link's capacity is divided into two logical channels.

Multiplexing

The link's capacity is divided into several logical channels. This allows for simultaneous transmission of multiple signals across a single physical link.



Multiplexing techniques

5 basic types of multiplexing techniques:

- ⌘ Space-Division multiplexing (SDM)
(not in Forouzan)
- ⌘ Frequency-Division Multiplexing (FDM)
- ⌘ Wavelength-Division Multiplexing (WDM)
(not in Kihl)
- ⌘ Time-Division Multiplexing (TDM)
- ⌘ Code-Division Multiplexing (CDM)

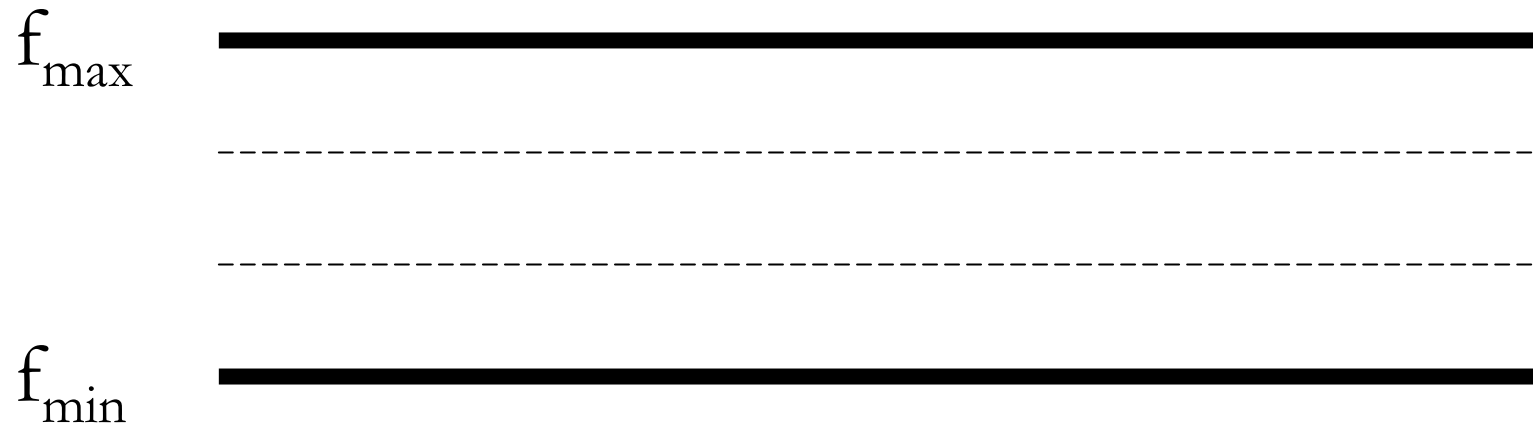
Space-Division Multiplexing (SDM)

SDM is used in fibre-optic cables. Each channel uses one optical fibre.



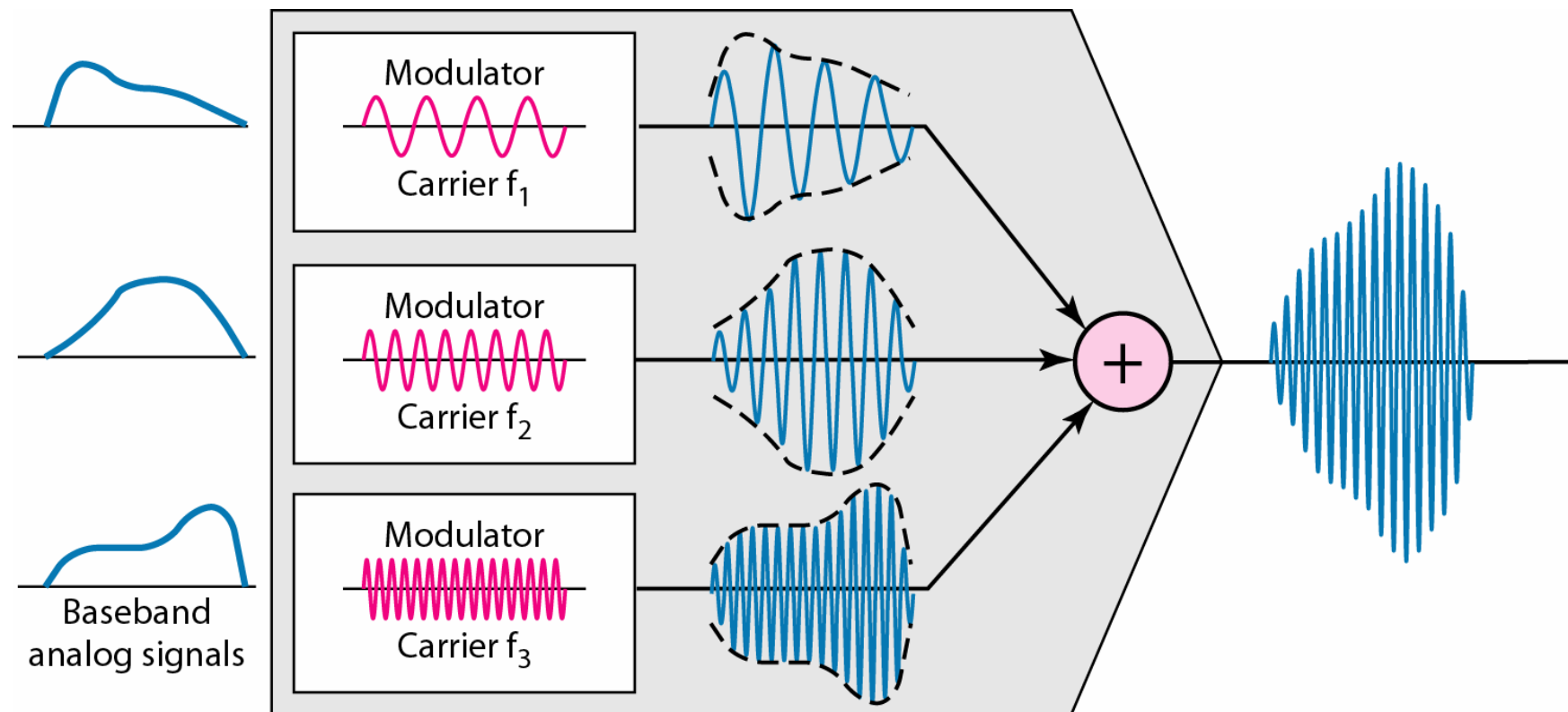
Frequency-Division Multiplexing (FDM)

FDM is an analog multiplexing technique that combines analog signals



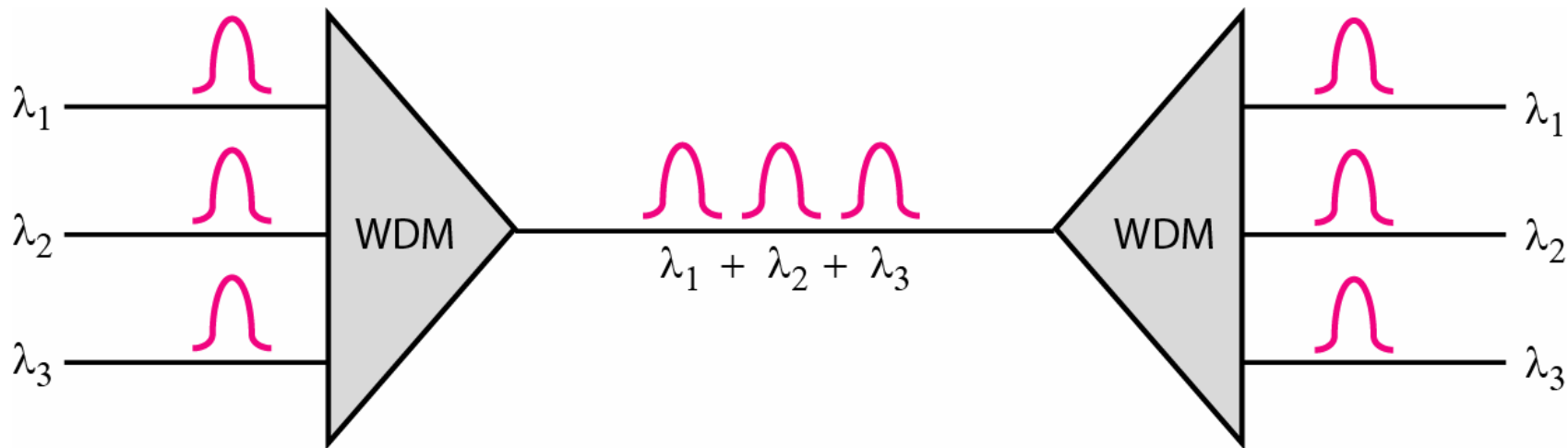
Each channel uses a unique carrier frequency.

FDM process



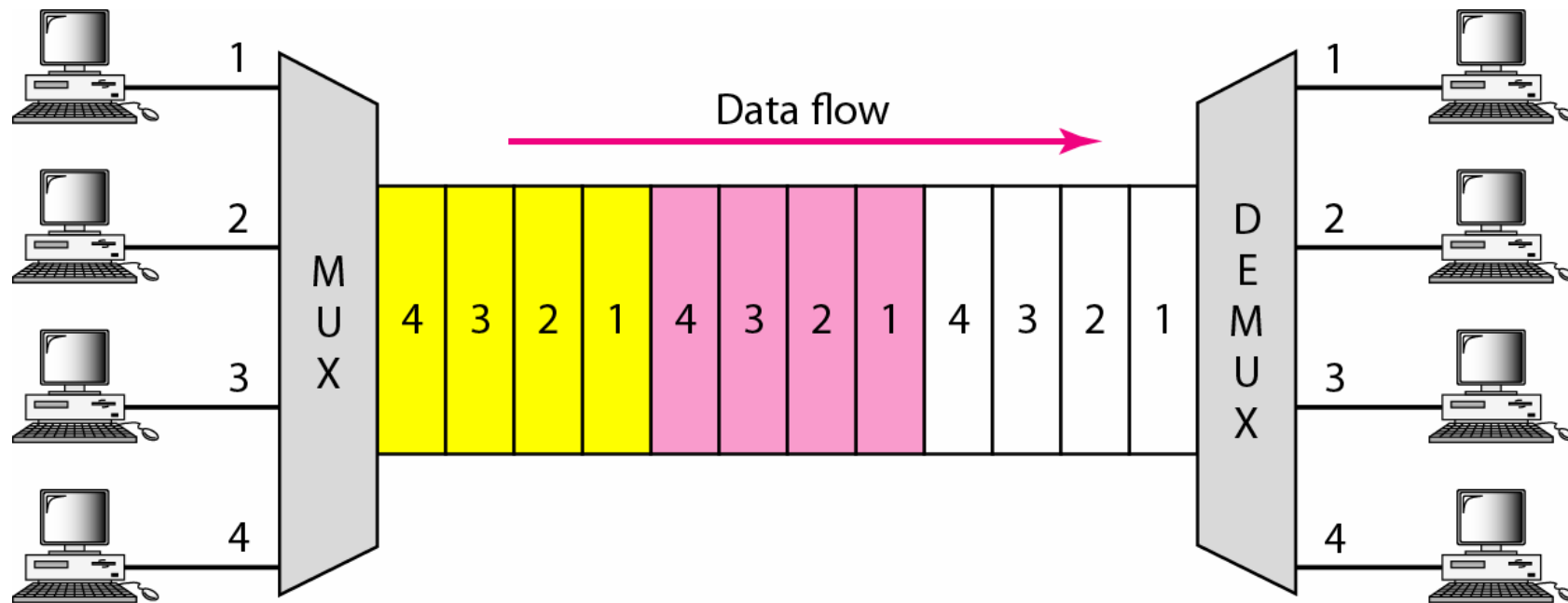
Wavelength-Division Multiplexing (WDM)

WDM is an analog multiplexing technique to combine optical signals.

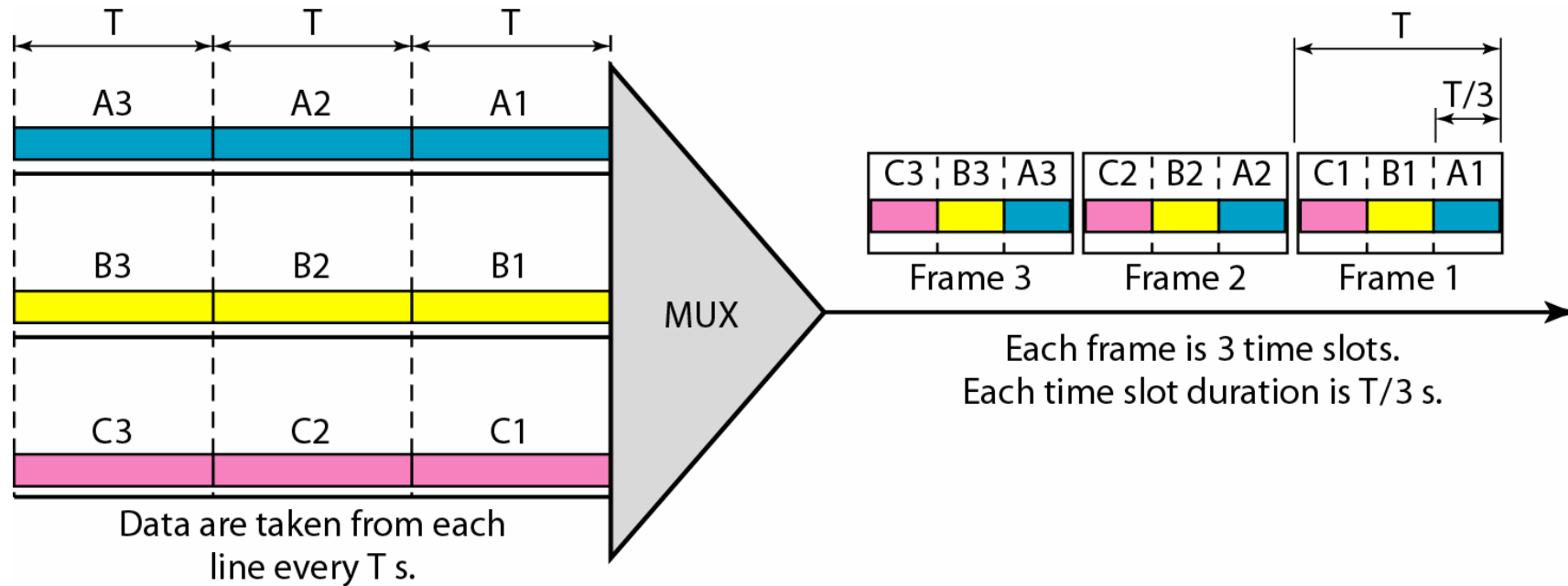


Time-Division Multiplexing (TDM)

In TDM, each channel occupies a portion of the **time** in the link.

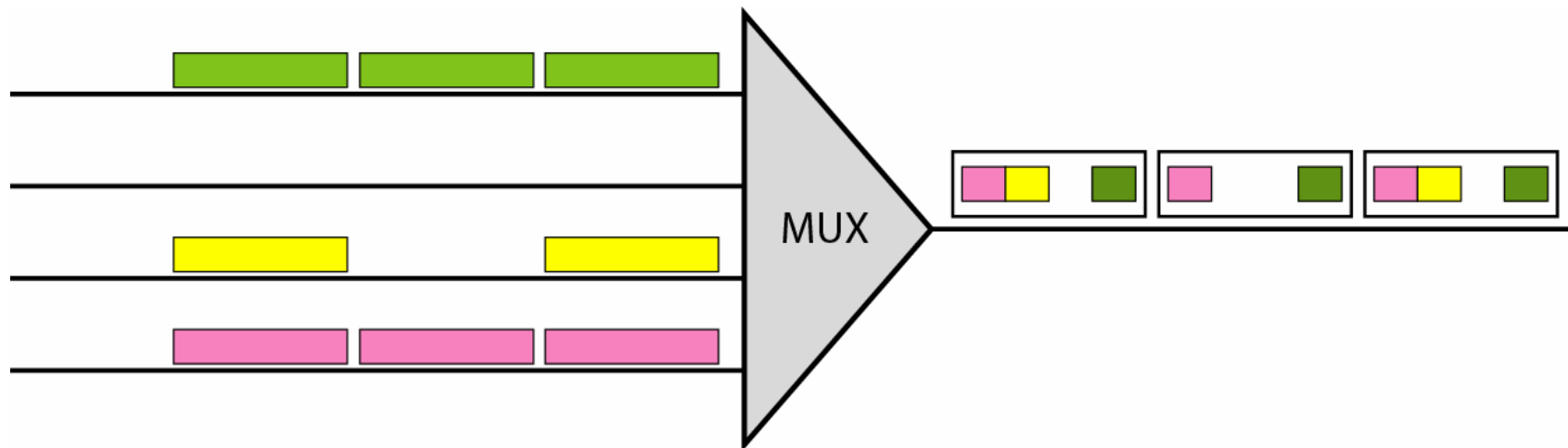


Synchronous TDM



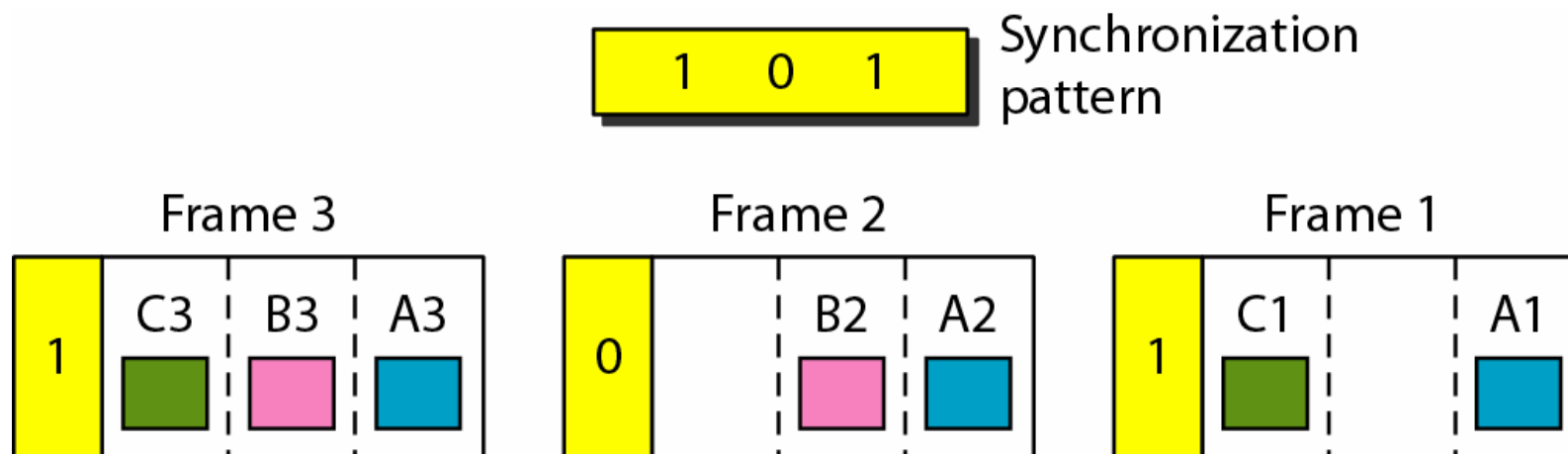
If a channel has nothing to send, its time slots will be empty!

Example: Empty slots



Frame synchronizing

If the multiplexor and the demultiplexor are not synchronized, bits may be received by the wrong channel. Therefore, synchronization bits (**framing bits**) are added in the beginning of each frame.



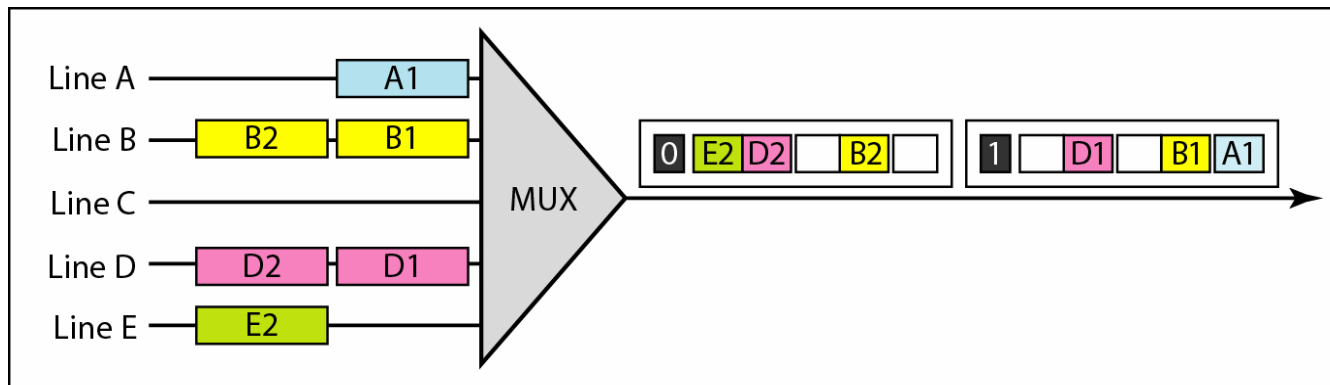
Statistical Time-Division Multiplexing

In Statistical TDM, the channels have no reserved time slots. Instead, slots are dynamically allocated.

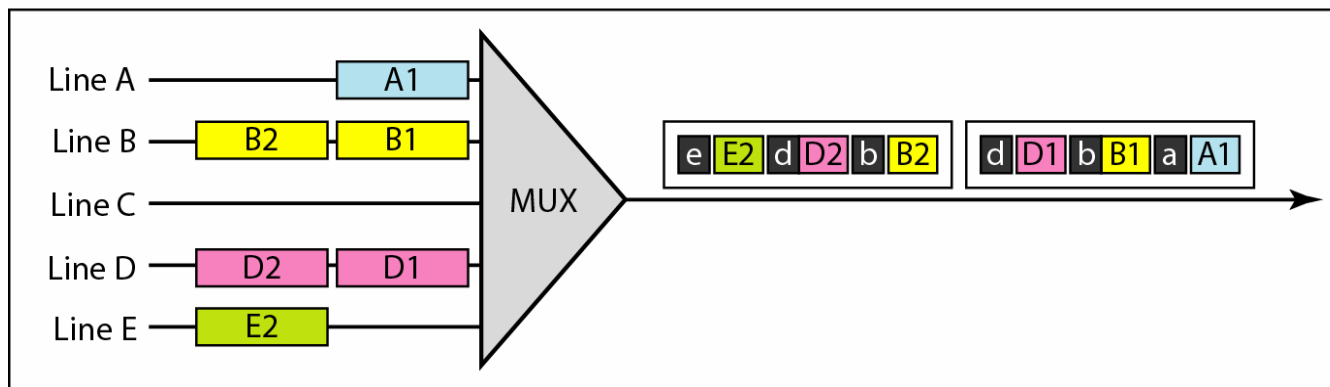
Data about the destination is added to each slot.

Statistical TDM is usually better than Synchronous TDM when not all channels transmit data all the time.

TDM comparison



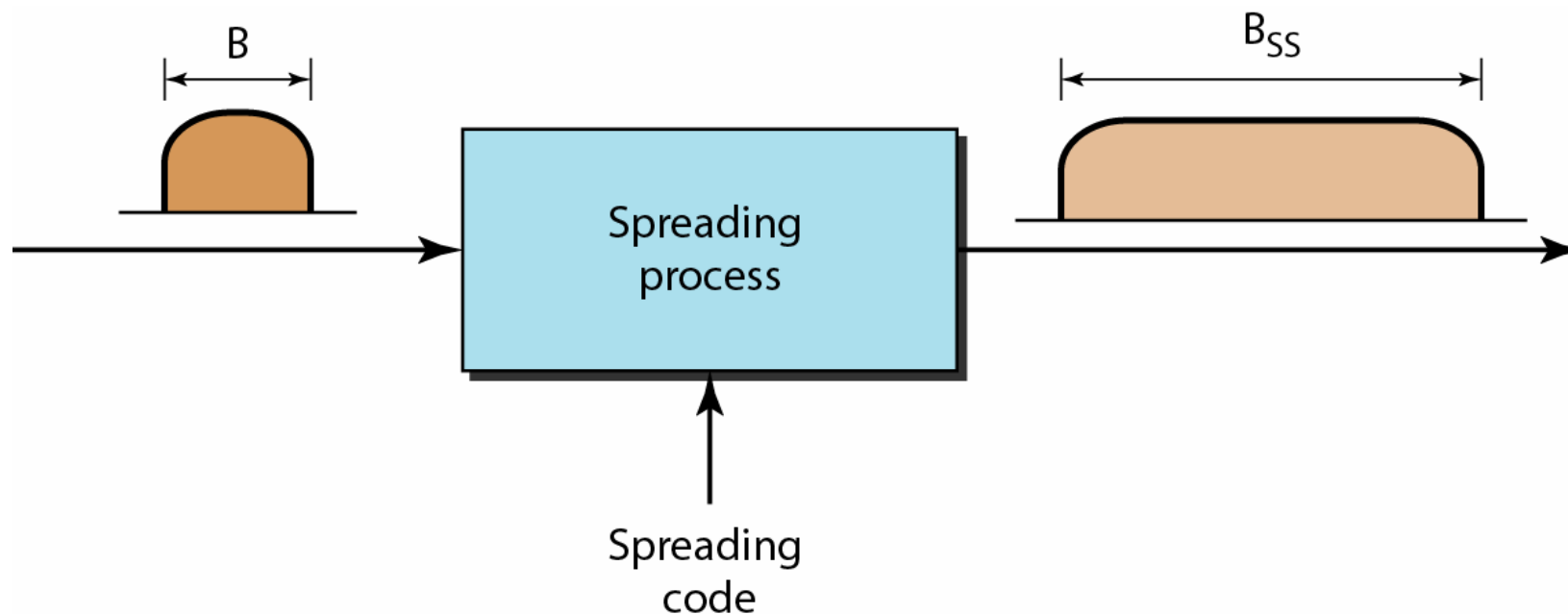
a. Synchronous TDM



b. Statistical TDM

Code-Division Multiplexing (Spread Spectrum)

CDM, or Spread Spectrum (SS), is a multiplexing technique for wireless links. The original signal is changed in a spreading process.

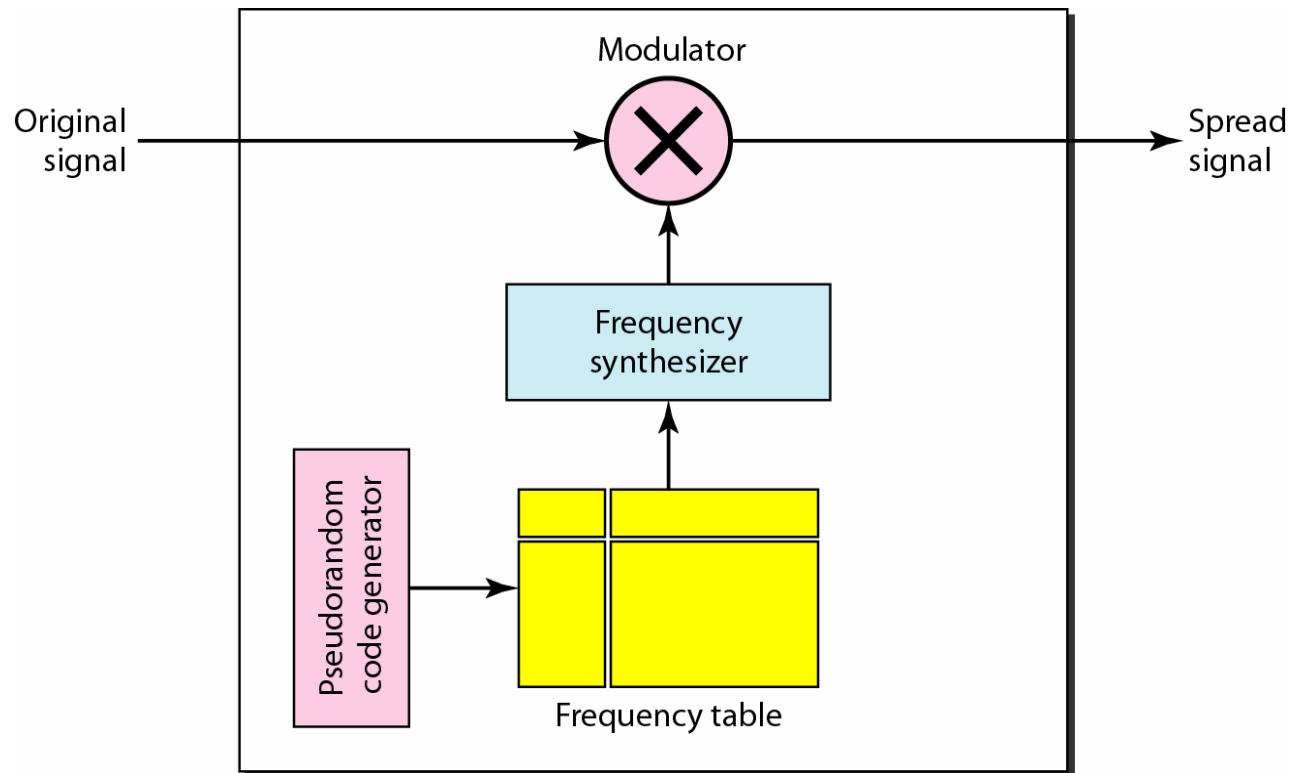


Spread Spectrum techniques

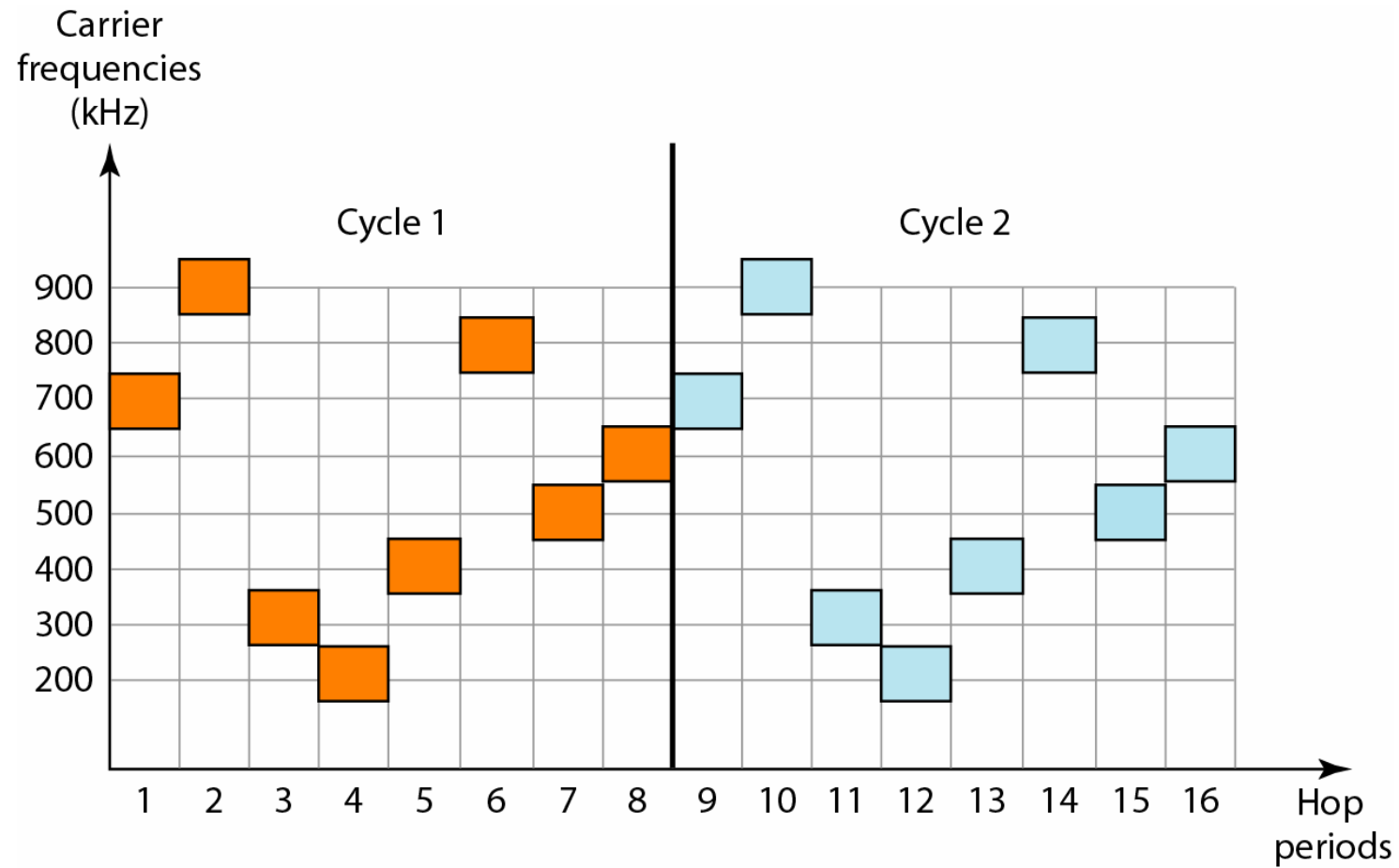
- Frequency Hopping Spread Spectrum (FHSS)
 - A source uses many carrier frequencies. One frequency is used at a time, but the frequencies are changed very often (e.g. 1000 times a second).
- Direct Sequence Spread Spectrum (DSSS)
 - Each data bit is replaced with n bits (called chips) using a unique spreading code. The code is chosen so that the combination of all other sources can be treated as white noise.

FHSS

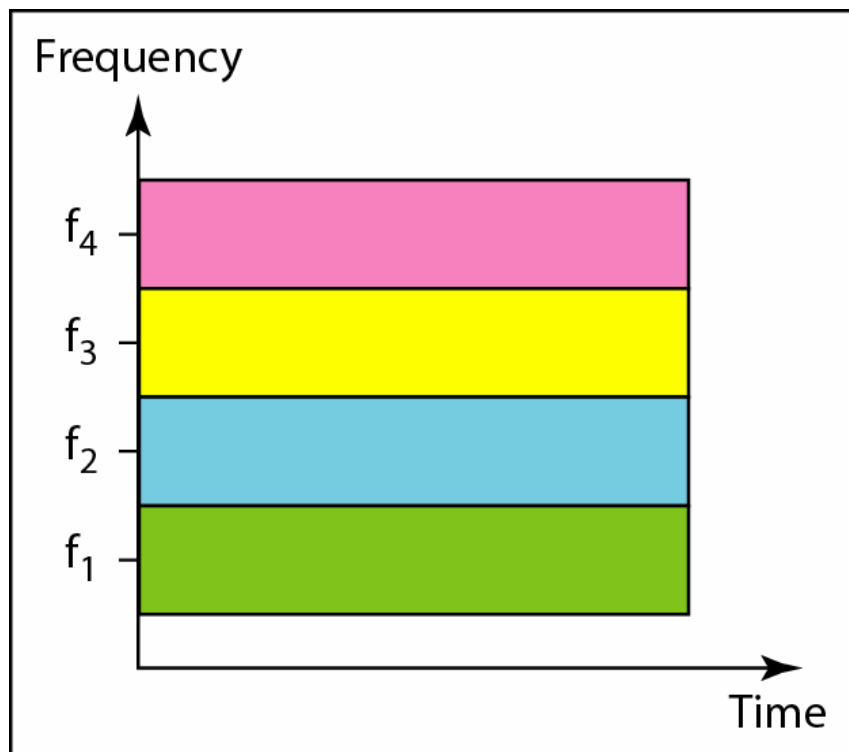
A source use a pseudorandom code generator to find the next carrier frequency.



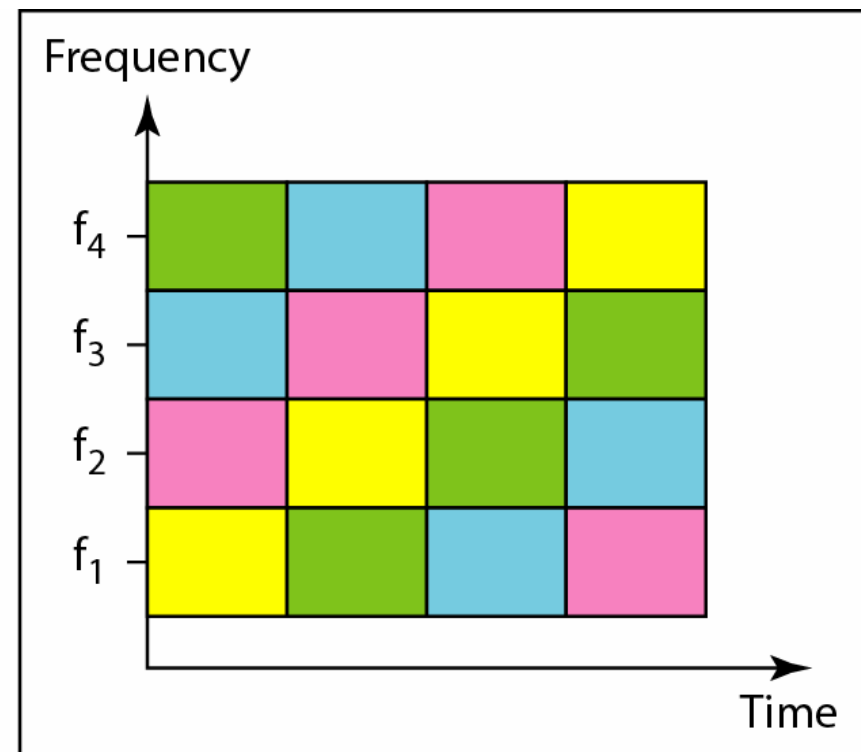
FHSS cycles



Comparison with FDM



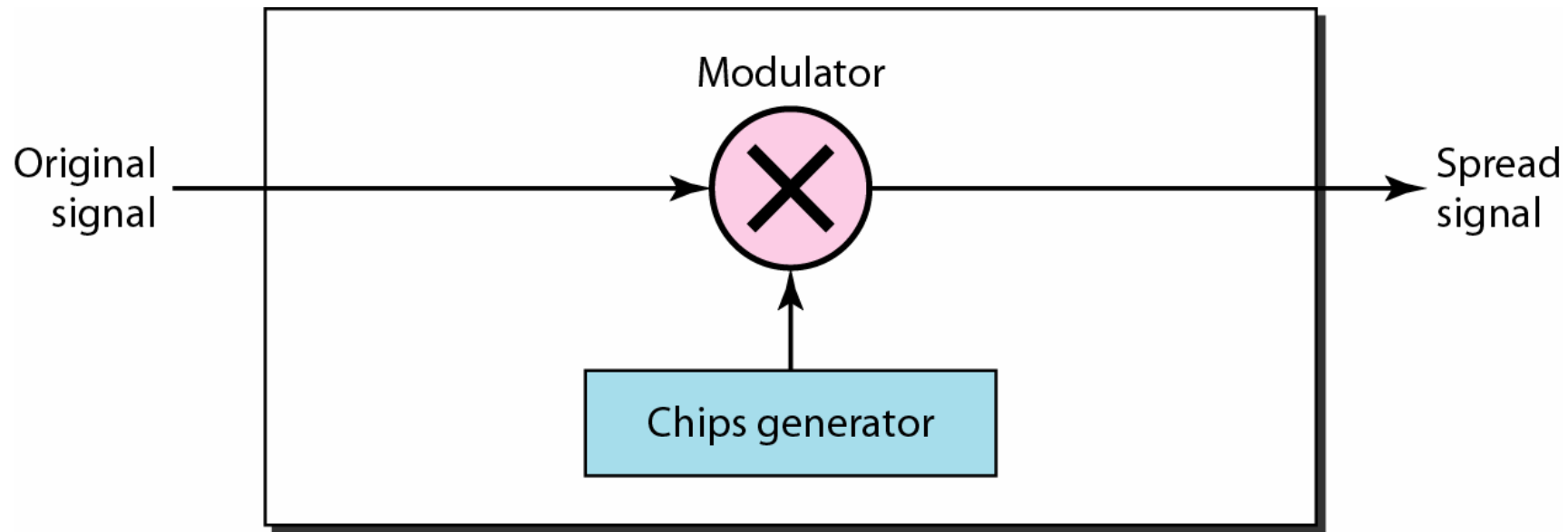
a. FDM



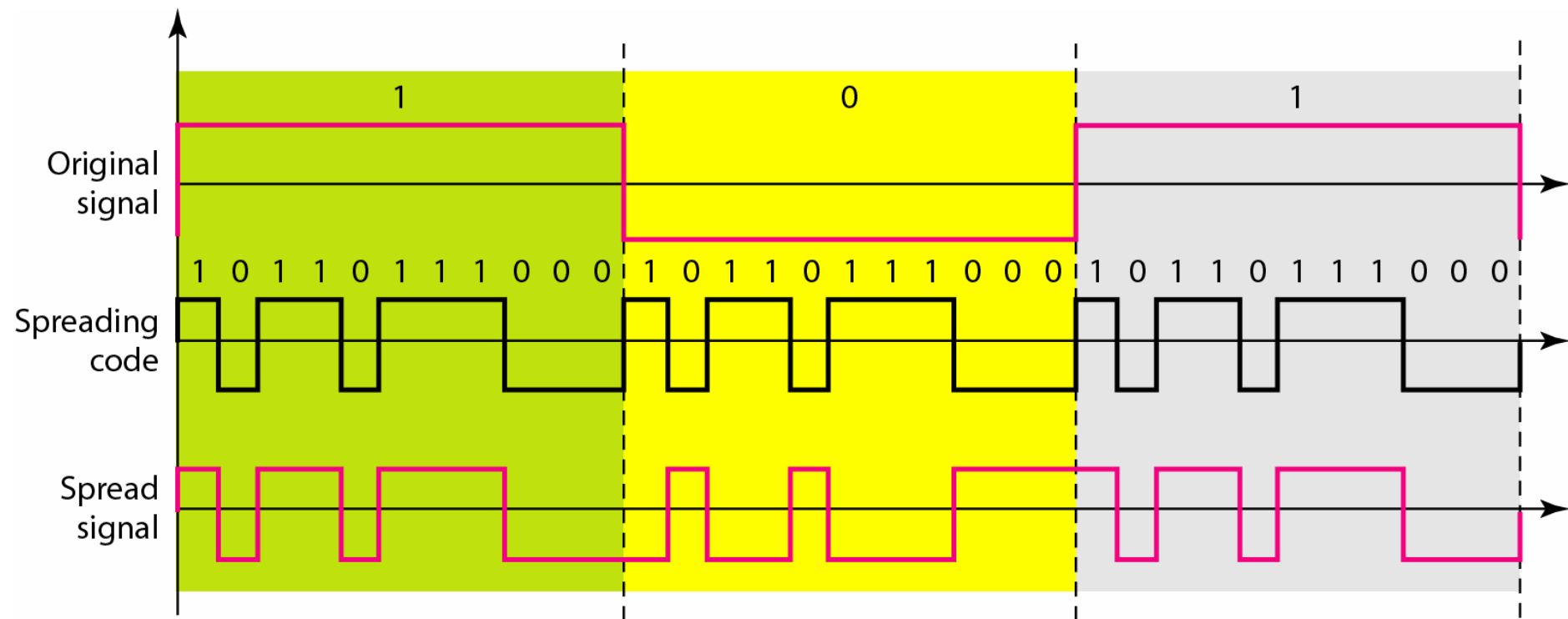
b. FHSS

DSSS

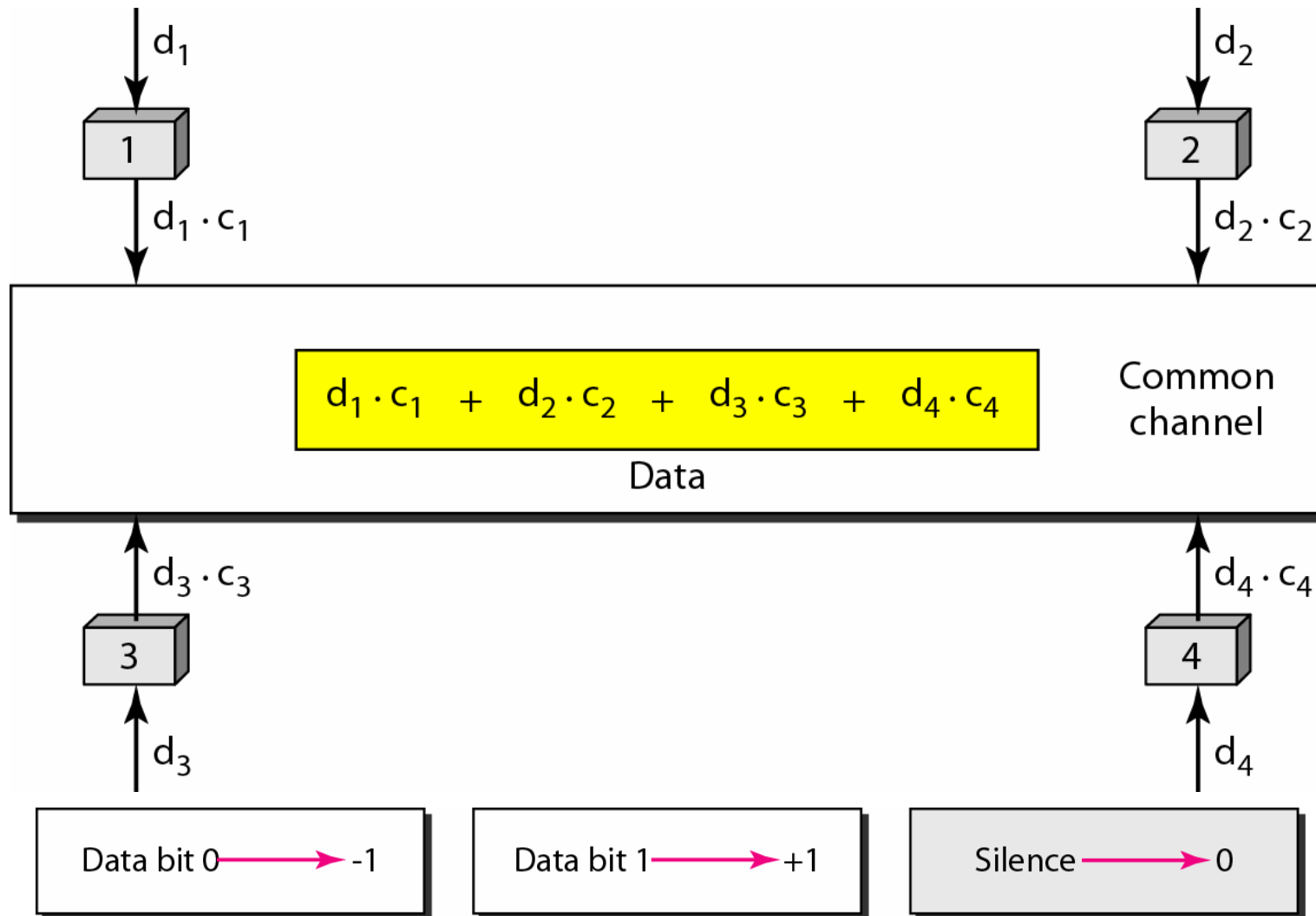
The original signal is combined with the spreading code.



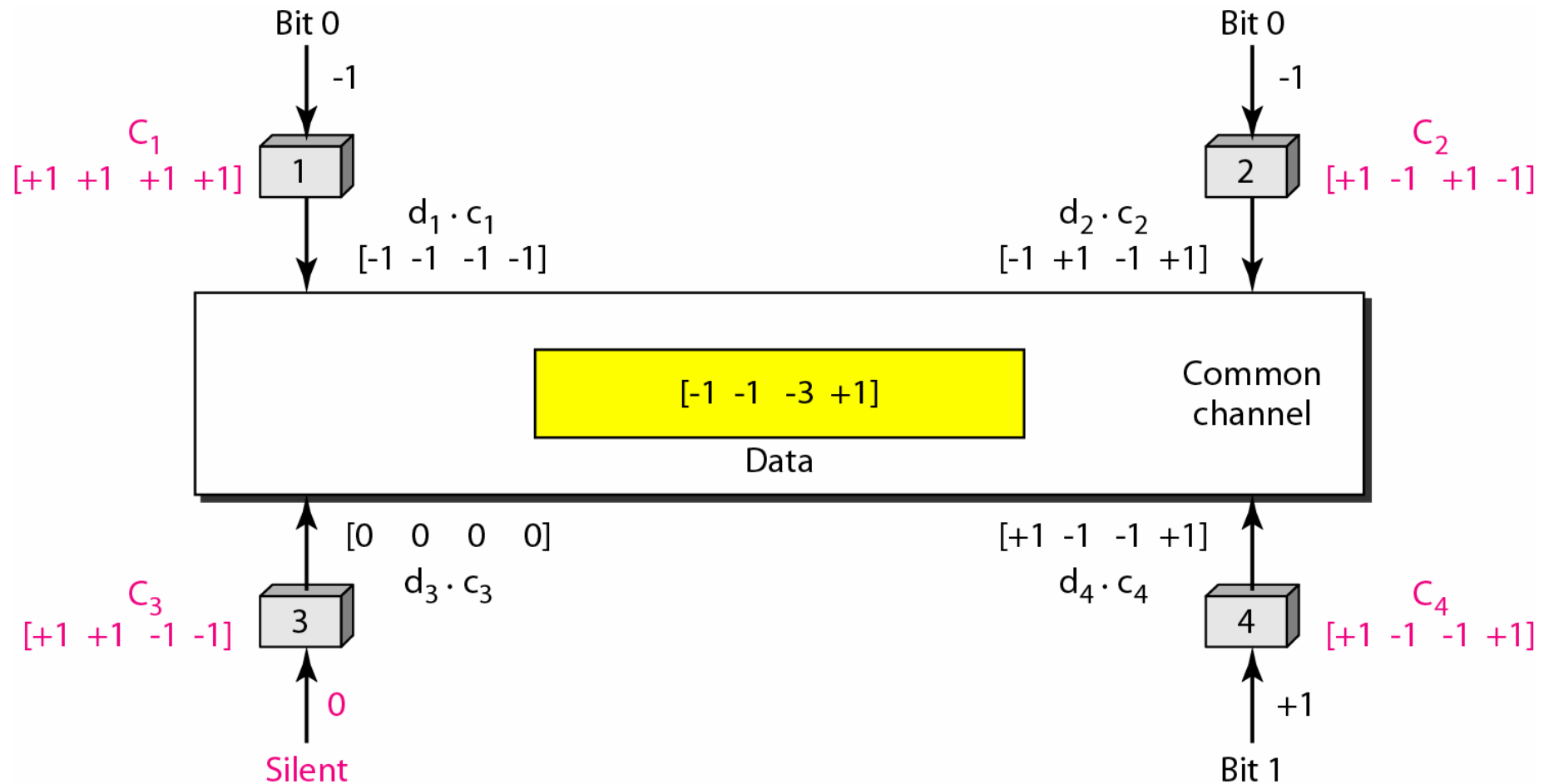
DSSS example



CDM example



CDM example (2)



CDM example (3)

